# Skills Summary

## Percent Applications — Rate, Whole, and Percent Change

*Use this reference while completing your worksheet or when studying.*

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**One relationship runs through all three skills:**

$$\frac{\text{part}}{\text{whole}} = \frac{\text{percent}}{100}$$

Two of the three are always given. **Name the missing one first** — that single decision is what tells you which skill below you are using. The part is the piece being described; the whole is what the percent is "of".

## Skill 1 — Percent rate from a part and a whole (percent-rate-from-parts)

**Missing: the percent.** Put the unknown on top of the 100: $\frac{\text{part}}{\text{whole}} = \frac{p}{100}$, so $p = \frac{\text{part}}{\text{whole}} \times 100$.

Shortcut: if the whole divides 100 evenly (25, 20, 50, 4, 5), scale the fraction to a denominator of 100 and read the percent straight off.

**Example:** 27 of 45 tickets were sold. What percent were sold?
**Step 1.** The part (27) and the whole (45) are both given and the percent is missing, so the unknown goes over 100.
**Step 2.** $\frac{27}{45} = \frac{p}{100} \Rightarrow 45p = 2700 \Rightarrow p = 60$. (Or simplify first: $\frac{27}{45} = \frac{3}{5} = \frac{60}{100}$.) $\Rightarrow$ **60%** of the tickets were sold

**Try it:** 51 of 60 seats are taken. What percent are taken?

check: **85%**

## Skill 2 — The whole from a part and a percent (whole-from-part-and-rate)

**Missing: the whole.** The unknown now sits in the *denominator*: $\dfrac{\text{part}}{n} = \dfrac{\text{percent}}{100}$, so $n = \dfrac{\text{part} \times 100}{\text{percent}}$.

**Size check:** when the percent is below 100, the whole must come out *bigger* than the part. If it does not, the two numbers were swapped.

**Example:** 24 students is 30% of the grade. How many students are in the grade?
**Step 1.** The percent (30) and the part (24) are given and the whole is missing, so the unknown belongs under the part.
**Step 2.** $\dfrac{24}{n} = \dfrac{30}{100} \Rightarrow 30n = 2400 \Rightarrow n = 80$. Check: $0.30 \times 80 = 24$ ✓ $\Rightarrow$ the grade has **80** students

**Try it:** 45 pages is 60% of a book. How many pages does the book have?

check: **75**

## Skill 3 — Percent increase or decrease (rate-from-change)

**Missing: the rate of a change.** The part is the *amount of change*; the whole is always the **original** amount:

$$\text{percent change} = \frac{|\text{new} - \text{original}|}{\text{original}} \times 100$$

Going up is an increase, going down a decrease — say which in your answer.

**Example:** A club grew from 80 members to 92. What percent increase is that?
**Step 1.** This compares a change to a starting amount, so the change is the part and the original 80 is the whole — not the new 92.
**Step 2.** Change $= 92 - 80 = 12$, so $\dfrac{12}{80} = \dfrac{15}{100}$. $\Rightarrow$ an increase of **15%**

**Try it:** A price fell from 150 dollars to 120 dollars. What percent decrease?

check: **20%**

**(!) Watch out:** (1) Dividing whole by part instead of part by whole gives a nonsense percent — 80 over 5.40 is over 1400%, so let the size of the answer warn you. (2) In a percent change, dividing by the *new* amount answers a different question. (3) After a 20% increase, the new total is 120% of the old one, so undo it by dividing by 1.20 — never by taking 20% off the new total.